

Feature Selection in Statistical Machine Learning

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Lecture 19

Recap and outline

- Feature attribution
- Local vs global attribution
- Shapley value framework

Feature attribution setup

Data: (p features and n samples)

$$(\{X_j^{(i)}\}_{j=1}^p, Y^{(i)})_{i=1}^n$$

Model/function:

$$\mathbb{E}[Y|X_1, X_2, \dots, X_p] = \sum_{j=1}^p \beta_j X_j$$

Prediction algorithm/estimator:

$$\widehat{f}(\cdot) = \mathcal{A}((\{X_j^{(i)}\}_{j=1}^p, Y^{(i)})_{i=1}^n) =$$

- For a *local* or *specific* instance/example with features $\tilde{X} = (\tilde{X}_1, \tilde{X}_2, \dots, \tilde{X}_p)$, prediction is $\widehat{f}(\tilde{X}_1, \tilde{X}_2, \dots, \tilde{X}_p)$
- **Question:** How much does each feature contribute to local prediction $\widehat{f}(\tilde{X}_1, \tilde{X}_2, \dots, \tilde{X}_p)$?

Feature attribution

- How do we assign an attribution ϕ_j for each feature $\tilde{X}_j$ that captures importance for the *individual prediction*?
- Key property (completeness/efficiency):

$$\sum_{j=1}^p \phi_j = \widehat{f}(\tilde{X}_1, \tilde{X}_2, \dots, \tilde{X}_p)$$

- Prediction can be decomposed into sum of p feature attributions
- Other properties often introduced

Examples

- Linear/logistic regression: $\phi_j = \widehat{\beta}_j \tilde{X}_j$
- General function: $\phi_j = \frac{\partial f}{\partial X_j} \tilde{X}_j$
- Can we develop a general framework for feature attribution?

Shapley value framework

- Axiomatic approach
- Originally developed in cooperative game theory (1951)
- Imagine you have p players and to you want to measure how much each player contributes to total utility
- Became prominent in machine learning/statistics only in the past 10 years or so because there was no general principled way to measure feature contribution
- In machine learning, players are our features ($X_1, X_2, \dots, X_p$)

Shapley value axioms

- General framework for feature attribution measures that satisfy 4 properties/axioms

1. Efficiency

$$\sum_{j=1}^p \phi_j = \widehat{f}(\tilde{X}_1, \tilde{X}_2, \dots, \tilde{X}_p)$$

2. Symmetry

If X_j and X_k are interchangeable then

$$\phi_j = \phi_k$$

3. Dummy

If X_k contains no information about Y

$$\phi_k = 0$$

4. Additivity

If $f = f_1 + f_2$, for all j

$$\phi_j(f) = \phi_j(f_1) + \phi_j(f_2)$$

Intuition of properties for feature attribution

1. Efficiency

$\sum_{j=1}^p \phi_j = \widehat{f}(\tilde{X}_1, \tilde{X}_2, \dots, \tilde{X}_p)$. As discussed last week, all feature contributions add up to overall prediction

2. Symmetry

If X_j and X_k are interchangeable then $\phi_j = \phi_k$. Two interchangeable features have the same contribution (equal credit assignment)

3. Dummy

If X_k contains no information about Y then $\phi_k = 0$. If X_k makes no (direct or indirect) contribution to prediction, attribution is 0

4. Additivity

If $f = f_1 + f_2$, for all j then $\phi_j(f) = \phi_j(f_1) + \phi_j(f_2)$. If f is a combination of two prediction functions, Shapley value is additive (useful for ensemble methods).

Shapley value general formula

Feature set $(X_1, X_2, \dots, X_p)$

$S \subset \{1, 2, \dots, p\}$ denotes any subset of indices

Value function:

$$v : 2^p \rightarrow \mathbb{R}$$

is a function over a subset of the features (will see various examples later)

Shapley value:

$$\phi_j(v) = \sum_{S \subset \{1, 2, \dots, p\} / \{j\}} \frac{|S|!(p - |S| - 1)!}{p!} (v(S \cup \{j\}) - v(S))$$

Provably satisfies all 4 properties

Shapley value:

$$\phi_j(v) = \sum_{S \subset \{1,2,\dots,p\}/\{j\}} \frac{|S|!(p-|S|-1)!}{p!} (v(S \cup \{j\}) - v(S))$$

- $(v(S \cup \{j\}) - v(S))$ is a of how much feature X_j improves utility when working with the features X_S
- $\frac{|S|!(p-|S|-1)!}{p!} = \frac{1}{p} \binom{p-1}{|S|}$ is a weight that takes accounts for the size of S relative to $p-1$

Shapley value ordering/ permutation and subset definition

Alternative and equivalent formula in terms of orderings of features:

$$\phi_j(v) = \frac{1}{p!} \sum_{\pi \in \Pi} (v(S^\pi \cup \{j\}) - v(S^\pi))$$

For each ordering of features $X_1, X_2, X_3, \dots, X_p$ defined as $\pi \in \Pi$, only include the features that are earlier in the ordering. Then take an average over all orderings

3-feature example

$$\Pi = (\pi_1, \pi_2, \pi_3, \pi_4, \pi_5, \pi_6)$$

For feature number X_1 :

$$\pi_1 = (1, 2, 3) \quad : \quad S^1 = \emptyset$$

$$\pi_2 = (1, 3, 2) \quad : \quad S^2 = \emptyset$$

$$\pi_3 = (2, 1, 3) \quad : \quad S^3 = \{2\}$$

$$\pi_4 = (3, 1, 2) \quad : \quad S^4 = \{3\}$$

$$\pi_5 = (2, 3, 1) \quad : \quad S^5 = \{2, 3\}$$

$$\pi_6 = (3, 2, 1) \quad : \quad S^6 = \{2, 3\}$$

$$\phi_1 \quad := \quad \frac{1}{6} \sum_{k=1}^6 (v(S^k \cup \{1\}) - v(S^k))$$

Example choices of $v(\cdot)$ functional

Conditional value:

$$v(S) = \mathbb{E}[f(X)|X_S = x_S]$$

Global risk (naturally connects to feature selection):

$$v(S) = \mathbb{E}[\ell(f_S(X_S), Y)]$$

Functional:

$$v(S) = f(x_S, x'_{S^c})$$

where x' is baseline/local point of interest

Functional (connected to integrated gradient):

$$v(S) = f(x' + 1_S \odot (x - x'))$$

where x' is baseline/local point of interest

In this course, we will mainly focus on first 2

Example 1: 2-feature linear model

$$Y = \beta_1 X_1 + \beta_2 X_2$$

where $\mathbb{E}[X_1] = \mathbb{E}[X_2] = 0$

$$v(S) = \mathbb{E}[f(X)|X_S = x_S]$$

$$v(\emptyset) = \mathbb{E}[\beta_1 X_1 + \beta_2 X_2] = 0$$

$$v(\{1\}) = \mathbb{E}[\beta_1 X_1 + \beta_2 X_2 | X_1 = x_1] = \beta_1 x_1 + \beta_2 \mathbb{E}[X_2 | X_1 = x_1]$$

$$v(\{2\}) = \mathbb{E}[\beta_1 X_1 + \beta_2 X_2 | X_2 = x_2] = \beta_1 \mathbb{E}[X_1 | X_2 = x_2] + \beta_2 x_2$$

$$v(\{1, 2\}) = \mathbb{E}[\beta_1 X_1 + \beta_2 X_2 | X_1 = x_1, X_2 = x_2] = \beta_1 x_1 + \beta_2 x_2$$

Example 1: Independent features

$$v(\emptyset) = \mathbb{E}[\beta_1 X_1 + \beta_2 X_2] = 0$$

$$v(\{1\}) = \mathbb{E}[\beta_1 X_1 + \beta_2 X_2 | X_1 = x_1] = \beta_1 x_1 + \beta_2 \mathbb{E}[X_2 | X_1 = x_1] = \beta_1 x_1$$

$$v(\{2\}) = \mathbb{E}[\beta_1 X_1 + \beta_2 X_2 | X_2 = x_2] = \beta_1 \mathbb{E}[X_1 | X_2 = x_2] + \beta_2 x_2 = \beta_2 x_2$$

$$v(\{1, 2\}) = \mathbb{E}[\beta_1 X_1 + \beta_2 X_2 | X_1 = x_1, X_2 = x_2] = \beta_1 x_1 + \beta_2 x_2$$

Following Shapley value calculation:

$$\phi_1 = \beta_1 x_1$$

and

$$\phi_2 = \beta_2 x_2$$

Example 1: Perfectly correlated features $X_1 = X_2$

$$v(\emptyset) = \mathbb{E}[\beta_1 X_1 + \beta_2 X_2] = 0$$

$$v(\{1\}) = \mathbb{E}[\beta_1 X_1 + \beta_2 X_2 | X_1 = x_1] = \beta_1 x_1 + \beta_2 \mathbb{E}[X_2 | X_1 = x_1] = (\beta_1 + \beta_2)x_1$$

$$v(\{2\}) = \mathbb{E}[\beta_1 X_1 + \beta_2 X_2 | X_2 = x_2] = \beta_1 \mathbb{E}[X_1 | X_2 = x_2] + \beta_2 x_2 = (\beta_1 + \beta_2)x_2$$

$$v(\{1, 2\}) = \mathbb{E}[\beta_1 X_1 + \beta_2 X_2 | X_1 = x_1, X_2 = x_2] = \beta_1 x_1 + \beta_2 x_2$$

Following Shapley value calculation:

$$\phi_1 = \frac{1}{2}(\beta_1 + \beta_2)x_1$$

and

$$\phi_2 = \frac{1}{2}(\beta_1 + \beta_2)x_1$$

Equal credit assignment principle

- In independent case, we get $\phi_j = \beta_j x_j$ as expected (what we used in previous lecture)
- In perfectly correlated case $X_1 = X_2$, $\phi_1 = \phi_2$
- Follows from symmetry property
- Equal credit assignment principle of Shapley values
- **Question:** How do feature selection methods we have seen so far handle perfectly correlated features (e.g. LOCO, Lasso, e.t.c.)?

Example 2: Pairwise interaction model

Interactions occur in practice. Recall negation example for language.
Consider model:

$$Y = X_1 \oplus X_2$$

where X_1, X_2 are independent coin flips.

- $\oplus$ is XOR meaning if $X_1 = X_2$, $Y = 0$ and if $X_1 \neq X_2$, $Y = 1$
- **Question:** What is the Shapley value with $v(S) = \mathbb{E}[f(X)|X_S = x_S]$

Example 2 cont.

$$v(\emptyset) = \mathbb{E}[X_1 \oplus X_2] = \frac{1}{2}$$

$$v(\{1\}) = \mathbb{E}[X_1 \oplus X_2 | X_1 = x_1] = \frac{1}{2}$$

$$v(\{2\}) = \mathbb{E}[X_1 \oplus X_2 | X_2 = x_2] = \frac{1}{2}$$

$$v(\{1, 2\}) = \mathbb{E}[X_1 \oplus X_2 | X_1 = x_1, X_2 = x_2] = x_1 \oplus x_2$$

Following Shapley value calculation:

$$\phi_1 = \frac{1}{2}(x_1 \oplus x_2 - \frac{1}{2})$$

and

$$\phi_2 = \frac{1}{2}(x_1 \oplus x_2 - \frac{1}{2})$$

Example 2 cont.

Shapley values:

$$\phi_1 = \frac{1}{2}(x_1 \oplus x_2 - \frac{1}{2})$$

and

$$\phi_2 = \frac{1}{2}(x_1 \oplus x_2 - \frac{1}{2})$$

- If $x_1 = x_2$, $\phi_1 = \phi_2 = -\frac{1}{4}$
- If $x_1 \neq x_2$, $\phi_1 = \phi_2 = \frac{1}{4}$

Equal credit assignment principle

- Once again notice that $\phi_1 = \phi_2$
- **Question:** How do feature selection methods we have seen so far handle perfectly correlated features (e.g. LOCO, Lasso, e.t.c.)?

Feature selection for interaction model

- Note the conditional independence statements $Y \perp X_1$, $Y \perp X_2$, $Y \perp X_1|X_2$, $Y \perp X_2|X_1$, $Y \not\perp \{X_1, X_2\}$
- Feature selection methods would select no features
- One of the weaknesses of feature selection methods is that they don't capture interaction effects well

Summary of results for two models

Recall some of the weaknesses/limitations of feature selection methods:

- Only capture global (not local) impacts
- Highly dependent/correlated features present identifiability challenges
- Interactions (like language model example) not usually captured

Shapley values "resolve" the issues:

- Local attribution (depends on (x_1, x_2) in all examples)
- For highly correlated features, equal credit assignment splits contributions evenly
- Same for interaction model

But....

- Shapley values address local vs global issue
- Claim that they address dependent features and interactions issue that feature selection methods struggle with (if you search LLMs or speak to most industry people, they would make this claim)
- However Shapley values really just imposing a different assumption (equal credit assignment) to feature selection methods
- Feature selection methods apply different principles (e.g. Lasso applies sparsity, LOCO picks neither, e.t.c.)
- In fact these two problems (highly correlated features or interactions) are essentially impossible to resolve using the idea of looking at influence of single features

To truly resolve these issues...

- Need to start looking at subsets of features
- For both examples (highly correlated features and interaction models)
 $Y \perp X_1|X_2$ and $Y \perp X_2|X_1$ but $Y \not\perp \{X_1, X_2\}$
- So it is clear there is a relationship when considering the pair $\{X_1, X_2\}$
- To completely resolve the issue need to consider all subsets of features (computationally expensive)
- Both Shapley values and feature selection have the limitation that they attempt to characterize effect/impact of one feature at a time